

Case Study 1: Unauthorized Nmap Network Scan

Analysis:-

The student used Nmap to scan a company's network without obtaining permission. Even though no system were locked and no data was stolen the scan was unauthorized. Ethical hackers must always obtain written permission before performing any security testing.

Ethical Analysis.

- Violates ethical hacking principles.
- No prior authorization from the company.
- Ethical hacking requires consent and defined scope.

Legal Analysis:-

Application Sections.

- Section 43 : Unauthorized access to Computer System
- Section 66 : Unauthorized access done dishonestly or fraudulently is punished.
- Section 72 :- Breach of Confidentiality and privacy.

Possible legal Consequences

- * Police Complaint
- * Define the scope of testing
- * Sign a legal agreement
- * perform only authorized scans
- * Report vulnerability confidentially.

Conclusion:- The students actions were unethical and Potentially illegal because permission are not obtained before scanning

Case Study 2:- SQL injection Vulnerability Disclosure

Analysis:- The Researcher discovered a SQL injection vulnerability but immediately posted the exploit and proof of concept online before informing the company attackers exploited the vulnerability and customers data was leaked.

Ethical Analysis:-

- * The Researcher acted irresponsibly.
- * Public disclosure before a fix increased the risks
- * Responsible disclosure protects users.

Legal Analysis:-

Section 43:- Unauthorized damage or access.

Section 66:- Computer related offence

Section 72:- Breach of confidentiality.

Section 66/66D:- it identifies theft.

Responsible Vulnerability disclosure.

- * Inform the organisation privately.
- * Provide technical details solely.
- * Give sufficient time to fix the issue.
- * Publicly disclose only after the vulnerability is patched.

Risks of Premature Disclosure

- * Hackers exploit the vulnerability
- * Customer data theft
- * Financial loss
- * Loss of Customer trust
- * Damage to Company Reputation.

Recommendation.

Research should always follow responsible vulnerability Disclosure (or) Co-ordinated Vulnerability Disclosure.

Conclusion:-

Publishing exploit details before informing the company was unethical and increased cyber risk.

Case study 3:- Phishing attack on employee

Types of cyber attacks

phishing:

Analysis: The employee received a fake email pretending to be from the IT Support team enlarged login credentials on a fake website the attackers used those credentials to access confidential company data

Legal Implications

Application sections

- * section 43: Unauthorized access
- * section 66: Computer-related offences.

Section 66C:- Identify theft

Section 66D:- Cheating by personation using computer resources.

Responsibilities of the organisation

- * Conduct employee awareness training.
- * Enable multi-factor Authentication MFA.
- * Use email security filters and spam
- * monitor security advisories and spam
- * enforce strong password policies.

Responsibility of the employee:

- * Verify suspicious emails before responding.
- * Never share passwords through email.
- * Report phishing attack or attempts immediately and follow company cybersecurity policies.

Preventive Measures:-

- * Regular phishing awareness programs.
- * MFA implementation.
- * Email filtering and anti phishing tools.
- * Regular response procedure.

Case study 4: Student exploits Pront vulnerability Before Posting.

Analysis:-

Although the student intended to report the vulnerability accessing customer information without authorization is illegal good intentions do not make unauthorized access lawful.

Ethical hacking vs Unauthorized hacking.

Ethical hacking

Written Permission obtained

Defined scope

Legal

helps improve security

Unauthorized hacking

No permission

illegal access

Violates laws

Confused to legal action

Relevant IT act, 2000:

Section 43: Unauthorized access and data extraction.

Section 66:- Computer related offences.

* Others.

Proper Reporting Process:

- * Do not exploit beyond minimal verification.
- * Collect only enough evidence to prove the vulnerability
- * Contact the bank through official channels
- * Maintain Confidentially.
- * Give the bank reasonable time to fix the issue.
- * Avoid public disclosure until the issue is resolved.

Conclusion:-

The students in ten times was ethical, but exploiting the vulnerability without authorization was not lawful. Ethical hackers must always obtain permission before testing systems.